

WRITING SAMPLE DISCLAIMER

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Lube Oil Filter Element — Removal and Installation

Engine Oil System / XR-700 Series Turboshaft Engine

Applicability: XR-700, XR-720,
XR-740

Schema: S1000D Issue 4.2

Language: EN-US (STE)

Category: Procedure

1 PRELIMINARY REQUIREMENTS

1.1 Required Conditions

#	Condition	Procedure Reference
1	Engine shutdown and secured against inadvertent start	DMC-XRENG-A-80-10-00-00A-710A-D
2	Engine oil temperature below 50°C (122°F)	—
3	Oil pressure system depressurized	DMC-XRENG-A-72-00-00-00A-521A-D

1.2 Support Equipment

Part Number	Description	Qty
T-72-601	Filter wrench, lube oil bowl	1
T-00-050	Torque wrench, 0–150 N·m (0–110 ft-lb)	1
T-00-091	Safety wire pliers	1
—	Drain pan, minimum 4-liter capacity	1

1.3 Expendable / Consumable Parts

Part Number	Description	Qty
XR-72-6001	Filter element, lube oil	1
AS3209-214	O-ring, filter bowl	1
CM-C-001	Cleaning solvent, approved	AR
CM-L-032	Lockwire, 0.032-in. diameter (304 CRES)	AR

2 PROCEDURE

2.1 REMOVAL

■ WARNING

Engine oil operates at high pressure and high temperature. Make sure the engine is shut down and the oil temperature is below 50°C (122°F) before you start this procedure. Hot oil under pressure can cause serious burns and injury.

CAUTION

Do not let oil contact electrical connectors or harness assemblies. Contamination can cause electrical faults and damage to components. Install protective caps on all open ports immediately after disconnection.

1. Put a drain pan under the lube oil filter assembly **(1)**.
2. Cut and remove the safety wire from the filter bowl **(2)**. Discard the used safety wire.
3. Use filter wrench (T-72-601) to loosen the filter bowl **(2)**. Turn the filter bowl counterclockwise to loosen it.
4. Remove the filter bowl **(2)** by hand. Let the oil drain into the drain pan.
5. Remove the filter element **(3)** from the filter bowl **(2)**.
6. Remove and discard the O-ring **(4)** from the groove in the filter bowl **(2)**.
7. Examine the filter element **(3)** for unusual metal particles or contamination.

NOTE

If you find unusual metal particles or contamination, do not continue this procedure. Report the finding to engineering. Follow the requirements in the Engine Condition Monitoring Program (DMC-XRENG-A-05-20-00-00A-040A-D).

8. Discard the used filter element **(3)**.

2.2 INSTALLATION

CAUTION

Do not reuse the O-ring. Always install a new O-ring (AS3209-214) during reassembly. A reused O-ring can cause oil leaks.

1. Clean the filter bowl **(2)** with approved cleaning solvent (CM-C-001). Make sure the bowl is clean and free of debris.
2. Apply a thin, even coat of clean engine oil to the new O-ring **(4)**.
3. Install the new O-ring **(4)** in the groove of the filter bowl **(2)**. Make sure the O-ring is seated evenly and is not twisted.
4. Install the new filter element **(3)** in the filter bowl **(2)**.

NOTE

Make sure the filter element **(3)** is fully seated in the filter bowl **(2)**. An incorrectly seated element can allow unfiltered oil to enter the lubrication system.

5. Apply a thin coat of clean engine oil to the threads of the filter bowl **(2)**.
6. Install the filter bowl **(2)** on the filter housing **(1)** by hand. Turn the bowl clockwise until it is hand-tight.
7. Use filter wrench (T-72-601) and torque wrench (T-00-050) to tighten the filter bowl **(2)**.

TORQUE	25–30 N·m (18–22 ft·lb)
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8. Install safety wire (CM-L-032) through the safety wire holes of the filter bowl **(2)** and filter housing **(1)**. Use safety wire pliers (T-00-091).

NOTE

Install the safety wire so that it holds the filter bowl **(2)** in the tightened position. Do not twist the wire more than 6 turns per inch.

9. Remove the drain pan from under the filter assembly.
10. Remove all tools, equipment, and consumable materials from the engine work area.

2.3 POST-INSTALLATION CHECK

1. Do the Engine Oil Level Check procedure before you start the engine.

Ref: DMC-XRENG-A-72-60-00-00A-700A-D

2. Start the engine and let it reach normal operating temperature.
3. Examine the filter bowl area for oil leaks.

■ WARNING

If you find oil leaks, shut the engine down immediately. Correct the cause of the leak before you continue. Do not operate the engine with an oil leak.

4. Shut down the engine.
 5. Do the Engine Oil Level Check procedure again and add oil as necessary.
- Ref: DMC-XRENG-A-72-60-00-00A-700A-D
6. Record the maintenance action in the aircraft maintenance log in accordance with applicable regulatory requirements.